

CSE482L/ETE334L

Project Report Writing Guideline

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North South University
[University Name]

Department of Electrical and Computer Engineering [Department
Name]

NSU Logo

Project Title:
[Write your project title here]

Course Code & Title: CSE482L:
[Course Title]

Section: [Section
Number]

Group Number:
[Section Number]

Group Members Name(s): [Full
Name(s)]

Group Members ID(s):
[ID(s)]

Faculty Name and Initial:
[Faculty Name]

Instructor:
[Instructor Name]

Submission Date:
[Date]

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(Abstract, Introduction, Problem Statement, Objectives, Proposed Solution, Target Users, Scope, System Architecture, Technology Stack, UI Design, Features, Database Design, Implementation, Performance & Security, Testing, Deployment, Limitations, Future Work, Conclusion, References)

Report Writing Guideline

This report must be written in formal academic language using clear and structured paragraphs. Each section should clearly explain the relevant part of the project. Avoid excessive bullet points; instead, use descriptive explanations. Diagrams, screenshots, and figures must be properly labeled and referenced within the text. Ensure consistency in formatting, headings, and terminology throughout the report.

Abstract / Executive Summary

Write a concise summary (150–250 words) describing the project. Begin with the problem context and explain the motivation behind developing the system. Clearly state the objectives and highlight the key features. Mention the technologies used, including frontend frameworks, backend technologies, databases, authentication systems, APIs, payment gateways, and AI integrations if applicable. Conclude with the expected outcome or impact.

1. Introduction

Introduce the project by explaining its purpose and background. Discuss the domain of the problem and its relevance in the modern context. Provide a brief overview of the system and its main functionalities.

2. Problem Statement

Clearly describe the problem your project aims to solve. Explain the limitations of existing systems and identify the gap your solution addresses. Justify why solving this problem is important.

3. Objectives

Present the objectives of the project in paragraph form. The objectives should include building a full-stack web application using modern frontend frameworks (React, Vue, or Angular), implementing backend services (Node.js with Express or Django), integrating databases, authentication systems, APIs, and payment solutions.

4. Proposed Solution

Describe the proposed system at a high level. Explain how the system works, its major components, and how it effectively solves the identified problem. Highlight any unique or innovative aspects.

5. Target Users

Identify the intended users of the system. Describe their roles, needs, and how they will interact with the application.

6. Scope of the Project

Define what features are included in the system and what is excluded. Clearly mention the boundaries and limitations of the implementation for this semester.

7. System Architecture

Describe the overall architecture of the system. Explain how the frontend, backend, and database interact. Discuss the data flow, API communication, and authentication process. Mention architectural patterns such as client-server or MVC.

8. Technology Stack

Describe and justify the technologies used in the project.

The frontend is developed using modern frameworks such as React, Vue, or Angular, along with CSS frameworks like Tailwind CSS or Bootstrap. The backend is implemented using Node.js with Express or Django. The database may include PostgreSQL, MongoDB, or Supabase.

Authentication is handled using services such as Firebase, Appwrite, Clerk, or JWT-based systems. External integrations include payment gateways and third-party APIs. If AI features are included, explain how AI models or APIs are used.

9. User Interface Design

Describe the design and layout of the system. Explain key pages such as the home page, login/register page, dashboard, and main functional interfaces. Discuss responsiveness and usability considerations.

10. Features and Functionalities

Explain the system features in detail.

Describe the authentication system, including user registration, login, and access control. Explain core functionalities such as CRUD operations, dashboards, and user interactions. If payment integration is included, describe the payment process and transaction handling.

Explain any external API integrations and how they enhance functionality. If AI features are included, describe their purpose and implementation.

11. Database Design and Schema

Describe the database structure used in the system. Explain entities or collections, their attributes, and relationships. Discuss how data is stored, retrieved, and managed.

12. Implementation

Explain how the system was developed. Describe how frontend and backend components are connected, how APIs are used, and how the database is integrated. Include relevant screenshots such as UI pages, authentication flow, dashboards, and system outputs with proper captions.

13. Performance and Security

Discuss system performance considerations such as loading speed, responsiveness, and optimization techniques. Explain the security measures implemented, including authentication, authorization, input validation, and protection against common vulnerabilities.

14. Testing

Describe how the system was tested. Explain different testing methods such as functional testing, usability testing, and API testing. Provide sample test cases and discuss results.

15. Deployment

Explain how the system is deployed. Describe the platforms used for hosting the frontend, backend, and database. Provide the live URL and explain how users can access the system.

16. Cost and Quotation

Explain how the system is deployed. Describe the platforms used for hosting the frontend, backend, and database. Provide the live URL and explain how users can access the system.

17. Limitations

Discuss the limitations of the current system. These may include missing features, performance issues, scalability limitations, or reliance on third-party services.

18. Future Work

Describe potential improvements and future enhancements. This may include adding advanced features, improving performance, integrating more AI capabilities, or expanding to mobile platforms.

19. Conclusion

Summarize the overall project. Reflect on how the objectives were achieved and evaluate the effectiveness of the solution. Briefly mention the learning outcomes.